An analyzer can have zero or more character filters. A character filter carries out the following specific functions:

- *Removes unwanted characters from an input stream*—For example, if the incoming text has HTML markup, as in "<h1>Where is my cheese?</h1>", the requirement is to drop the `<h1>` and `</h1>` tags.
- *Adds to or replaces additional characters in the existing stream*—If the input field contains 0s and 1s, we may want to replace them with "false" and "true", respectively. Or if the input stream includes the character `?`, we can map it to the word "beta" and index the field.

We can use three character filters to construct an analyzer: the `html_strip`, `mapping`, and `pattern` filters. We saw these in action earlier, and in this section, we briefly review the semantics.

7.6.1 HTML strip (hmtl_strip) filter

As the name suggests, this filter strips unwanted HTML tags from the input fields. For example, when an input field with a value of `<h1>Where is my cheese?</h1>` is processed by the `html_strip` character filters, the `<h1>` and `</h1>` tags are purged, leaving "Where is my cheese?". We can test the `html_strip` character filter using the `_analyze` API.

Listing 7.34 The `html_strip` character filter in action

```
POST _analyze
{
  "text":"<h1>Where is my cheese?</h1>",
  "tokenizer": "standard",
  "char_filter": ["html_strip"]
}
```

The character filter simply strips the HTML tags from the input field and produces the tokens "Where", "is", "my", and "Cheese". However, we may need to avoid parsing the input field for certain HTML tags. For example, the business requirement may be to strip `<h1>` tags from the following input but preserve the preformatted (`<pre>`) tags:

```
<h1>Where is my cheese?</h1>
<pre>We are human beings who look out for cheese constantly!</pre>
```

We can configure the `html_strip` filter to add an `escaped_tags` array with a list of tags to leave unparsed. The first step is to create an index with the required custom analyzer.

Listing 7.35 Custom analyzer with an added filter configuration

```
PUT index_with_html_strip_filter
{
  "settings": {
    "analysis": {
      "analyzer": {
```